

HAT-P-32b

R-1342

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-32_200919 · created 2026-07-31T21:44:35+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2459111.7706 +/- 0.0012 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1786 +/- 0.0063
Transit depth (Rp/Rs)^2	3.19 +/- 0.23 [%]
Orbital Inclination (inc)	84.03 +/- 1.16
Airmass coefficient 1 (a1)	1.091 +/- 0.013
Airmass coefficient 2 (a2)	-0.0772 +/- 0.0057
Scatter in the residuals of the lightcurve fit is	1.21 %
Best Comparison Star	#2 - [570, 414]
Optimal Aperture	3.44
Optimal Annulus	11.37
Transit Duration (day)	0.1223 +/- 0.0047

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.1786 $\pm$ 0.0063 vs 0.1489 (deviation 4.3 σ)
Duration fit vs published	0.1223 d vs 0.1304 d (deviation 1.58 σ)
Mid-time O-C	8.3 min
Depth SNR	25.9
Residual scatter	1.21 %

Light curve

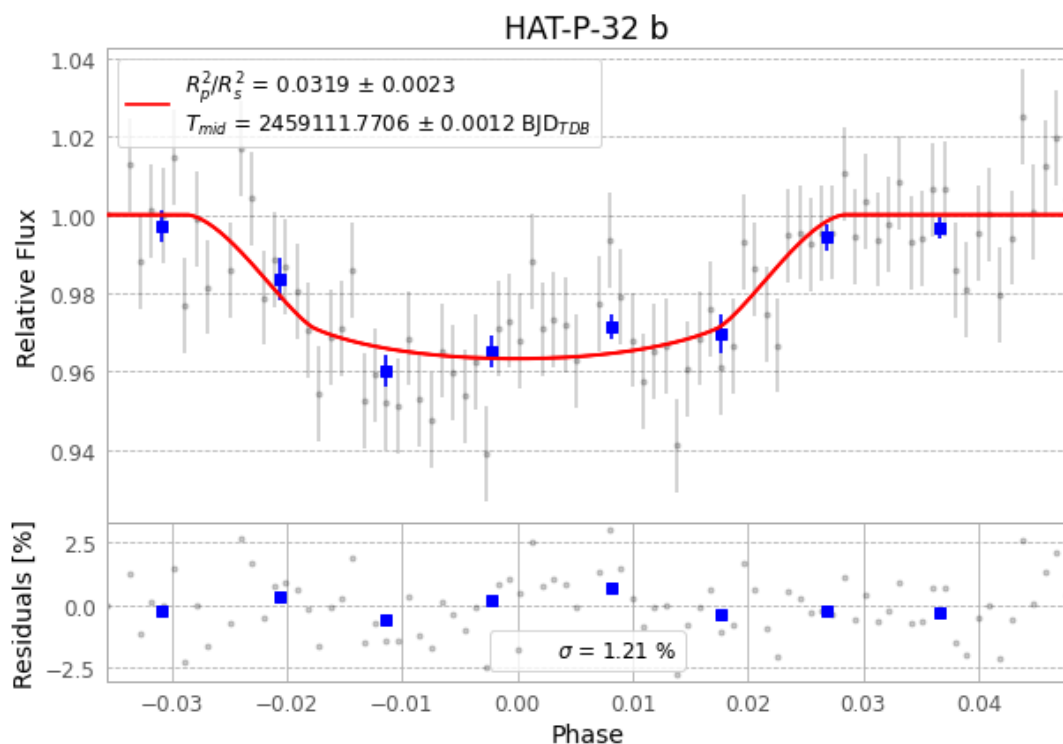


Figure 1 — FinalLightCurve_HAT-P-32 b_2020-09-18.png (SHA-256 d7a3c1ba9314c9eb...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [498, 278], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 3766008e6bb9caae...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

85 FITS frames (56 MB), HATP-32200919043312.FITS → HATP-32200919085112.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-32-200919.log (ca72905a3842...), FinalLightCurve_HAT-P-32 b_2020-09-18.png (d7a3c1ba9314...), FinalLightCurve_HAT-P-32 b_2020-09-18.csv (5b2d0f128d75...)

Compute resources

Wall time 150.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 21:44Z.